Multi-modal Editing of Synthesized Speech: An Empirical Study with ProsoLab

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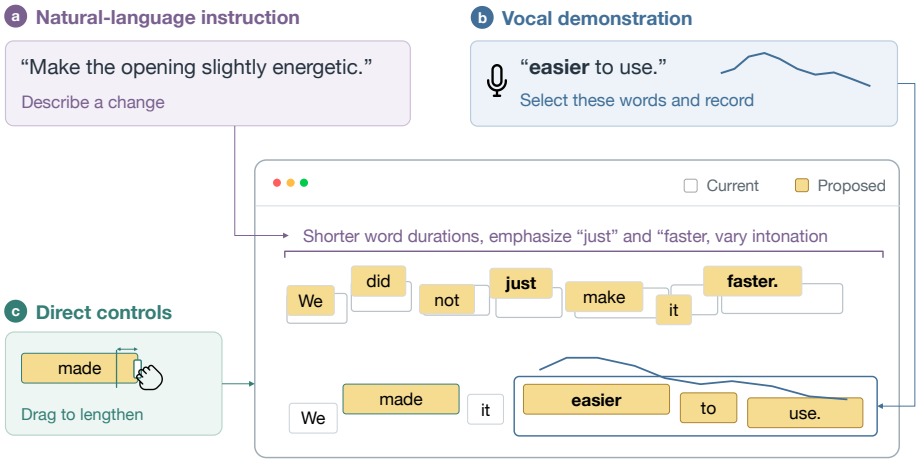


Fig. 1. ProsoLab supports three input methods for editing speech prosody: (a) Natural-language instruction interprets high-level intent (e.g., "Make the opening slightly energetic") into multi-parameter acoustic changes; (b) Vocal demonstration captures user voice input to transfer acoustic properties to selected target words; and (c) Direct controls enable fine-grained visual manipulation of word duration, pauses, and pitch on the spatial canvas. The canvas displays proposed edits (yellow blocks) overlaid on baseline contours (white blocks).

People use text-to-speech systems to generate narration from written scripts, yet they may recognize problems in the generated delivery without knowing how to express the changes they want. We present ProsoLab, a speech editor that integrates natural-language instruction, vocal demonstration, and direct controls through a global-to-local canvas interface. We conducted a two-part study with twelve participants, comparing the three methods on assigned tasks and examining how participants combined them during open-ended editing. Natural-language instruction and vocal demonstration were more often used to initiate edits, whereas direct controls were used as the final method for 67.4% of edited sentences to further refine details while retaining earlier changes. Participants switched methods to repair results, explore alternatives, and address new subgoals revealed by intermediate outputs. These findings show that the three speech-editing methods can serve complementary roles across successive revisions, motivating interfaces that support transitions between methods and preserve useful earlier decisions.

CCS Concepts: • **Human-centered computing** → **Interactive systems and tools**; **Empirical studies in HCI**.

Additional Key Words and Phrases: Speech Editing, Multi-modal Input

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1 Introduction

Content creators often use text-to-speech (TTS) systems to produce narration for scripted media, such as presentation videos, without the need to record every line themselves. A common workflow begins with a script and a selected voice. The creator listens to the generated speech and revises the sentence whose delivery does not fit the intended presentation. Current production tools support this process through local regeneration and controls over delivery [10]. However, a suitable voice and correctly spoken words do not ensure that the narration communicates the intended message. For example, a creator reporting that a system is "three times faster" may want to emphasize "three times" so that the improvement stands out, then pause before explaining its significance. After the speech is generated, the creator still needs to guide the expression of the speech content and judge whether the revisions have achieved the expected goals.

Existing interfaces offer different ways to guide the expressions. Natural-language instructions allow users to describe how speech should sound without specifying acoustic parameters [3]. Direct controls instead let users adjust acoustic parameters, such as pitch and speed [4]. Vocal demonstration lets users provide an example of the intended delivery using their own speech [10, 17, 25]. Instead of describing a change or setting a value, a user can record the same sentence to guide the TTS system in generating speech with the recorded delivery. These methods place different demands on users. An instruction may leave room for interpretation regarding the scope of the change, while a direct adjustment requires the user to select a control and an amount. Demonstration provides a performed example, but requires the user to produce the desired delivery. These differences matter when an edit shifts from shaping the overall delivery to refining specific words, and the user must decide how to achieve it. Although prior systems establish these editing capabilities, evidence remains limited about how the three methods compare across tasks with different scopes and prosodic goals, and what interaction support users need to use them together. We therefore ask: How do the three methods support different speech-editing tasks, and how do users choose and combine them during open-ended editing?

To inform the design of an interface for investigating these questions, we conducted a formative study with six non-native English-speaking academics who had created narrated presentation videos, which identified needs regarding both delivery and revision. Participants wanted important content to stand out and pacing to reflect the structure of their message. When reviewing revisions, participants wanted to locate changes and compare their effects. Participants reported having partly forgotten the earlier rendering, and expected visual indications of changes alongside audio previews. These arguments suggest that speech editing needs to support specifying changes in separate selectable acoustic properties and checking the effects of those changes.

To address these editing needs and study the three methods within one workflow, we developed ProsoLab, a speech editor that integrates natural-language instructions, direct controls, and vocal demonstrations through a global-to-local canvas. We chose emphasis and pitch controls together with word and pause durations to support changes in delivery at selected locations. Users can adjust these properties directly on the canvas or propose changes through instructions and recorded demonstrations. All three methods operate on the same speech, and accepted edits are retained when the user switches methods. To meet the comparison needs, the interface displays proposed changes alongside the current speech representation and provides audio previews before acceptance.

To understand how users perform speech-editing tasks with each method and combine methods across revisions, we designed a two-part within-subjects study with ProsoLab. The first part compared the three methods on assigned tasks that combine local prominence or timing goals with broader expressive goals. We examined participants' judgments of goal achievement and control alongside editing time, iteration count, and workload. The second part allowed participants to choose and combine methods while editing a complete paragraph. We examined how editing goals inform initial method choice, what prompts switching, and the roles of different methods across successive revisions. Questionnaires and interviews also address participants' experience of the integrated interface and their use of visual and audio feedback when judging edits.

In assigned tasks, direct controls received higher perceived-control ratings than the other two methods. Natural-language instruction required the most editing time, while vocal demonstration involved the fewest recorded editing cycles and direct controls involved the most. Participants found direct controls useful when they knew which local property to adjust and by how much. Natural-language instructions allowed them to describe a broad direction, while vocal demonstrations helped them express rhythm without deciding parameter values.

During open-ended editing, a method's contribution extended beyond whether it supplied the final edit or was used most frequently. Direct controls supplied the last accepted edit for 64 of 95 edited sentences (67.4%), yet each of the three methods was identified by four participants as contributing most to their final results. After using natural-language instruction to establish an overall style or vocal demonstration to illustrate delivery, direct controls could then be used to refine selected details while retaining earlier changes. Switching was not limited to repairing unsatisfactory results: some participants tried another method after obtaining an acceptable version to explore alternatives, or identified a further editing goal while listening.

In summary, we make the following contributions:

- ProsoLab, a system integrating vocal demonstration, natural-language instruction, and direct controls with interface support to facilitate speech-editing workflows.
- A two-part lab study of to compare the three methods on different editing tasks and how users select, switch between, and combine them during continued revision in open ended tasks.

2 Related Work

2.1 Interaction Methods for Editing Speech Delivery

Editing speech delivery changes how a script is spoken rather than what it says. Table 1 compares representative approaches, distinguishing the input capabilities of synthesis models from the interactions and evaluations of speech-editing interfaces. Prior systems support this process through three main interaction methods. Users can describe an intended change in natural language, provide speech that contains the intended delivery, or directly adjust selected speech parameters. These methods differ in what users can express, the level of control they provide, and the feedback available after an edit.

Natural-language instruction allows users to describe a desired delivery without identifying the acoustic values. The instruction-based models support requests for expressive changes and iterative modifications, and the evaluation includes instruction following and the quality of the generated speech [15, 44]. SpeakEasy brings such requests into an authoring interface alongside sentence-speed control. Participants rated its speech as better matching their personal standards than speech produced with the comparison tools, and reported no significant difference in the level of effort between them [3]. This links instruction-based control to the user's judgment of the editing process and its output.

Direct control lets users specify changes to selected speech parameters. Synthesis models support local adjustments through acoustic values or drawn contours [5, 24, 41]. Interactive systems expose these parameters through word-level representations, including movable word blocks and detailed parameter editors [4, 13]. These representations connect an adjustment to its location in the script. However, Voicesetting's evaluation found that detailed control required more editing effort even when users valued the resulting speech. Therefore, it is necessary to consider the greater control power with the workload required to use that control power.

Vocal demonstration specifies delivery through the user's performance. Prosody-transfer models use reference speech to guide generated delivery while seeking to preserve the target speaker's identity [18, 33]. This approach also appears in editing interfaces: VoCo uses recorded exemplars to guide local resynthesis, while ElevenLabs Actor Mode and Murf Say It My Way use recorded performances to guide generated speech [10, 17, 25]. Recorded delivery can also become directly editable by editing the word in the script. [2, 28].

The methods are not mutually exclusive. However, their availability does not determine which method is best suited for a particular editor. Interface comparisons in SpeakEasy and Voicesetting examine different tool configurations rather than isolating the three methods on matched tasks [3, 13]. We developed ProsoLab as a general editing environment and designed a comparison across task scope and prosodic dimensions. This comparison examines how each method supports the intended result and how much control and effort users experience when making the edit.

2.2 Expressing Intent and Evaluating Speech Changes

Speech editing requires users to connect a communicative goal to an available action and then judge whether the resulting delivery achieves that goal. These demands correspond to the gulfs of execution and evaluation [16]. Hutchins et al. further distinguish semantic distance from articulatory distance [16]. In speech editing, moving a word up by direct controls may clearly indicate a higher pitch, but it's uncertain whether this change will make the sentence sound more confident. Natural-language input allows users to express their goals in a familiar way, but familiarity does not guarantee consistent understanding. Research on prompting identifies difficulties in specifying effective instructions and anticipating the outputs they will produce [34, 45]. For speech, an instruction such as "sound more confident" describes an intended impression while leaving its acoustic interpretation to the system. Conversely, explicit parameter changes can identify a specific operation, but users need to judge for themselves whether it will produce the expected effect. Vocal demonstration offers a possibility: a user who can perform a desired delivery can provide an example without specifying its acoustic components. The translation demand then shifts toward producing the intended delivery and deciding which aspects of the recording should guide the edit.

Representations other than language can help users express aspects of a goal that are difficult to describe. Canvas-like interfaces let users combine or position visible representations to steer concepts without relying entirely on verbal descriptions [6, 7]. Besides, research on external representations explains how persistent objects support comparisons and reasoning that users can remember [19]. Research on generative AI also shows how inspectable intermediate results can support diagnosis and intervention [36, 43]. In speech editing, observing where changes occur and listening to their effects can provide different information for evaluation.

This perspective informs the feedback in ProsoLab. Its word-aligned representation provides a common reference for locating proposed changes and comparing them with the current speech, while audio previews support judgments of the resulting delivery. This representation aims to link editing operations to their intended effects rather than to encode the full communicative meaning of speech. We will study how users interpret this feedback and use that information to decide what changes to make next.

Table 1. Representative approaches to speech delivery and voice attribute control. Symbols distinguish input forms, not performance or completeness. Model inputs do not imply an interactive editing interface.

	Delivery-control inputs				
System or model	**Instruction**	**Vocal**	**Direct control**	**Scope**	**Evaluation focus**
Interactive systems and commercial tools					
VoCo [17]	–	○	Parameters	Word/phrase	Output and expert tests
TypeTalker [2]	–	–	Pause markers	Inter-word pauses	Interface comparison
Voicesetting [13]	–	–	Parameters	Token/span	Interface comparison
Prosodic Marionette [4]	–	–	Word blocks	Words and pauses	Prosodic production tasks
SpeakEasy [3]	●	–	Speed slider	Script/sentence	Interface and workflow study
ElevenLabs Studio [10, 11]	○	●	Sliders; timing	Selection/paragraph	–
Murf Studio [25, 26]	–	●	Graph + sliders	Word/sentence	–
Synthesis and editing models					
InstructSpeech [15]	●	–	–	Utterance	Instruction and output tests
VoxEditor [32]	●	–	Edit strength	Utterance	Attribute and output tests
Step-Audio-EditX [44]	●	–	–	Utterance; local events	Repeated-edit benchmarks
Ctrl-P [24]	–	–	Parameters	Phone to utterance	Control and listening tests
DrawSpeech [5]	–	–	Contours	Local contours	Sketch and output tests
SpeechPalette [41]	–	–	Pitch/duration	Word/segment	Automated benchmarks
CopyCat [18]	–	●	–	Within utterance	Transfer and output tests
ProsoLab (this work)	●	●	Word blocks	Word to paragraph	Method comparison; switch strategy

Language: ● free-form instructions; ○ inline audio tags.
Vocal: ● user-supplied target performance (recorded, uploaded, or selected); ○ reference use reported, but a user-supplied target is not established.
– Not reported (NR) for the defined role. Simply retaining the source speech time, speaker identity reference, and training audio is insufficient to constitute a vocal demonstration.

2.3 Method Choice and Transitions in Iterative Editing

An editing method's role can change as users experience its results. Studies of multimodal interaction show that users select methods according to task demands rather than using every available modality [27]. Their choices also depend on the method's efficiency and the input's performance [31]. DesignPrompt describes how users adjust their goals and strategies as they inspect intermediate outputs [29]. Unsuccessful attempts provide one reason to change methods. Multimodal correction research has found that switching input methods can improve recovery after speech recognition errors and that users can learn to avoid ineffective correction methods [35]. Therefore, the use of methods needs to be understood in conjunction with the current goal and whether previous interactions were successful.

Generative-editing interfaces support this iterative work through different forms of control. DynaVis turns a language request into a persistent control for subsequent adjustments [38]. PromptCharm similarly combines prompt revision with controls to adjust model attention and regenerate selected image regions [42]. Together with creative-coding interfaces that connect language-based exploration to editable code [1], these systems illustrate how a generated result can remain available for further work through another method.

Combining methods also requires attention to what each method can do. For example, multimodal video editing and image generation use language to specify an operation and sketching to identify its spatial target [30, 37, 39]. Similarly, DirectGPT lets users target generated objects through direct manipulation and enter instructions via natural language [23]. CARE distinguishes equivalence, where alternative methods can each achieve a goal, from complementarity, where inputs must be combined to achieve it [9]. It also distinguishes between system capabilities and user preferences for different approaches to different aspects of a task. In this way, the editing goals and the contribution of each operation become important units of analysis, rather than treating each switch as evidence of complementarity.

For speech delivery, a user might demonstrate a sentence's rhythm and then adjust one pause, or turn to a recording after an instruction produces an unsatisfactory result. Our open-ended paragraph-editing study with ProsoLab is designed to examine initial choices and subsequent transitions in relation to editing goals and intermediate results. CARE provides a basis for interpreting method relations, while participants' behavior and narratives are used to investigate when and why they allocate work among different methods.

3 Formative Study

We conducted a formative study to understand the difficulties non-native English-speaking academics face when producing narrated presentation videos and to identify their needs for a speech-editing interface. The study was designed to identify both the problems participants encountered in their prior production workflows and how they addressed them when listening to AI-generated speech. The findings informed three design goals for the proposed system.

3.1 Participants, Procedure, and Data Analysis

Participants. Six participants (P1–P6) were recruited from a research institute. All participants identified as non-native English speakers and had produced at least one narrated presentation video for academic purposes, such as conference submissions, demo videos, or course assignments, within the 12 months before the study. Each session lasted approximately one hour.

Procedure. Each session was conducted as a semi-structured interview organized into two phases. In the first phase, participants were asked to describe a past text-to-speech experience in which they were asked what was most frustrating or time-consuming, how they decided between recording their own voice and using a text-to-speech system, and what problems they encountered when using either approach. In the second phase, participants listened to two AI-generated renderings of one of their own presentation scripts. Using each participant's own scripts ensured that their responses were grounded in real communicative goals rather than in reactions to unfamiliar material. The first rendering was produced using ElevenLabs v3 without additional prosodic tags. Participants assessed whether the delivery matched their intentions and described the changes they wanted. We provided their script and unmodified instructions to Gemini 3.1 Pro, which generated and inserted prosodic tags into the script. Researchers did not rewrite participants' instructions or manually insert the tags. The tagged script was then synthesized using ElevenLabs v3 to produce the second rendering. Participants compared the renderings and discussed any remaining differences from their intended delivery.

Data Analysis. All sessions were audio-recorded and transcribed. One author analyzed the interview transcripts through qualitative coding and affinity grouping. The author selected sentences related to the interview questions and labeled them. The labeled excerpts are placed on the canvas and grouped according to the similarity of the experiences they describe or the editing needs they require. These groups were used to develop the themes reported below.

3.2 Findings

3.2.1 Participants Wanted Delivery to Reflect Content Structure and Intent. Participants wanted speech delivery to reflect which parts of their content mattered most. P5 wanted greater emphasis on "dynamically analyze" because it described a core feature of the work, but felt that the generated speech gave it similar emphasis to the other items in the sentence. P1 wanted emphasis on words that expressed an attitude toward the content, while P3 points out a specific statement about a chart that should be stronger. These requirements link the prominence of the selected content to what the participant wants the audience to understand.

Participants also used pacing and pauses to direct attention and separate ideas. P2 and P4 wanted less important phrases delivered faster and key information delivered more slowly. Similarly, P6 described reading background explanations faster while slowing down for important figures shown on the slides: "background content read faster, and key figures slower." P1 wanted more separation between sentences, while P2 wanted pauses of different lengths to indicate where listeners should pay attention. Therefore, to achieve the ideal rhythm, adjustments need to be made to word reading speed and the pauses between words.

Participants used intonation to express the emotion of the speech. P2 associated consistently low, downward sentence endings with an unwanted negative impression and felt that consistent intonation made the delivery sound artificial. P4 wanted a more fluctuating and less formal delivery to convey a dilemma. These descriptions clarify broader expression goals, always targeting complete sentences or longer paragraphs rather than specific words or phrases.

3.2.2 Participants Wanted to Identify and Compare Changes Across Versions. Comparing a revised speech with the earlier version can be difficult, even when a participant notices a preferred change. When asked to compare the two renderings, P5 reported having partly forgotten the previous version, so they needed to listen to the previous version more times to notice the revised emphasis on "dynamically." P5 requested a more direct comparison and proposed highlighting differences between versions to identify changes more quickly.

When asked how the changes should be presented, participants described feedback that would connect proposed changes to their location. P2 thought reviewing the speech solely via audio playback was too slow and wanted a visual overview before deciding what to preview. Pre-defined symbols could indicate the type of change, allowing the user to "roughly understand what changes have been made before choosing to listen." When deciding whether a change is appropriate, it is also important to understand what exactly it changes. P6 requested explanations of why changes were proposed and what benefits they were intended to provide. Together, these thoughts expressed a preference for locating and understanding changes alongside listening to their effects.

3.3 Summary of Editing Needs

The findings inform what users need to change and inspect. As a design response, we chose emphasis control to adjust prominence and pitch control to support intonation changes. Word duration supports local pacing adjustments, while pause duration controls separation between words or sentences. These four editable properties provide a focused set of controls for the delivery goals participants described.

The need for comparison informed our decision to display proposed changes alongside the current speech in a word-aligned representation, with audio previews to check their effects. Drawing on related work and our research questions, we connect vocal demonstration, natural-language instruction, and direct manipulation to this representation, thereby forming a concrete editing environment.

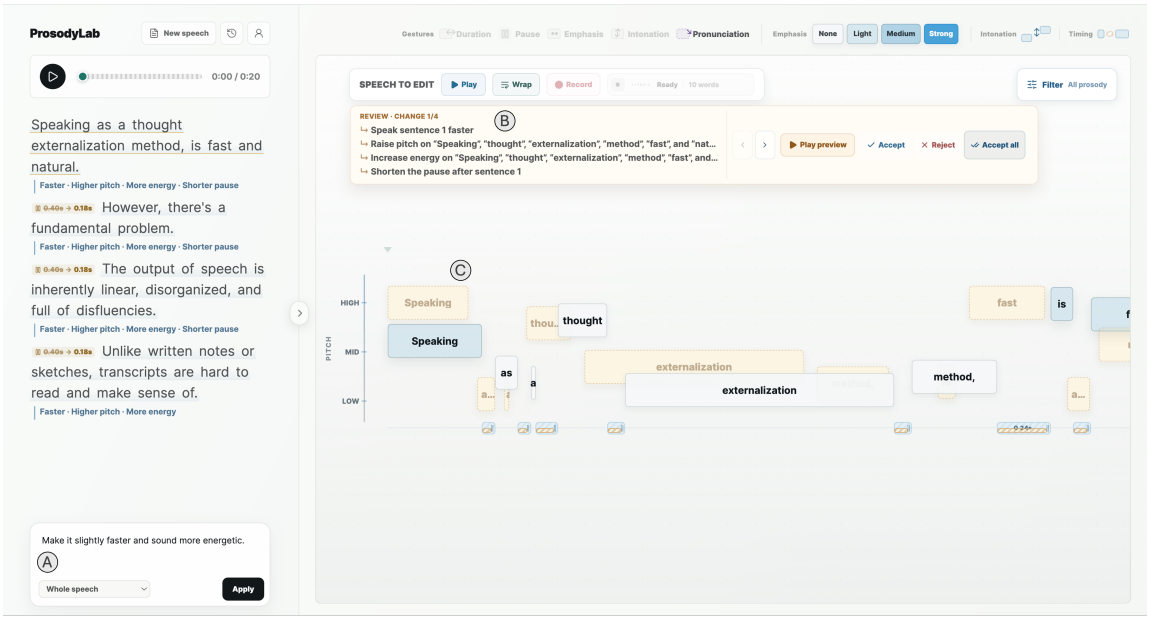


Fig. 2. The natural-language instruction workflow. (A) High-level prompting: The user inputs an open-ended descriptive instruction (e.g., "Make it slightly faster and sound more energetic") and selects the target scope ("Whole speech"). (B) Parameterized review card: The system decomposes the prompt into concrete acoustic adjustments across pitch, energy, speech rate, and pauses for each sentence. The proposed changes are also shown in the transcript, so users can see an overview before opening each sentence's editing panel. (C) Visual preview: Proposed edits (yellow blocks) are overlaid against the original words to visualize relative shifts in pitch level, word duration, and timing across the sentence. The user can play an audio preview, choose to Accept, Reject, or Accept all.

4 ProsoLab

Drawing on the findings of the formative study and the design goals, we developed ProsoLab, a speech-editing tool that supports vocal demonstration, natural-language instruction, and direct parameter adjustment. In this section, we will first describe an example scenario with ProsoLab (which does not represent any participant's workflow), then illustrate how we build its key features.

4.1 Example Scenario

Consider Alex, who is preparing a CHI presentation video. He enters a four-sentence script into ProsoLab and generates the narration. To engage the audience, he inputs the instruction, "Make it slightly faster and sound more energetic." The proposed changes are displayed as a yellow overlay on the original speech, with explanations alongside the canvas. He reads the explanations and compares the preview with the original before accepting the changes (Figure 2).

When introducing the benefits of speaking, Alex wants "is fast and natural" to stand out. He selects these four word areas and records only that phrase with his intended delivery. His recorded delivery appears as a yellow overlay on the selected span (Figure 3A). He listens to the generated preview and clicks "Apply preview" to transfer the delivery to those words (Figure 3B and C). In the same sentence, he notices that "externalization" is not pronounced as he intends. He selects the word, records its pronunciation, and listens to the pronunciation candidates before applying his

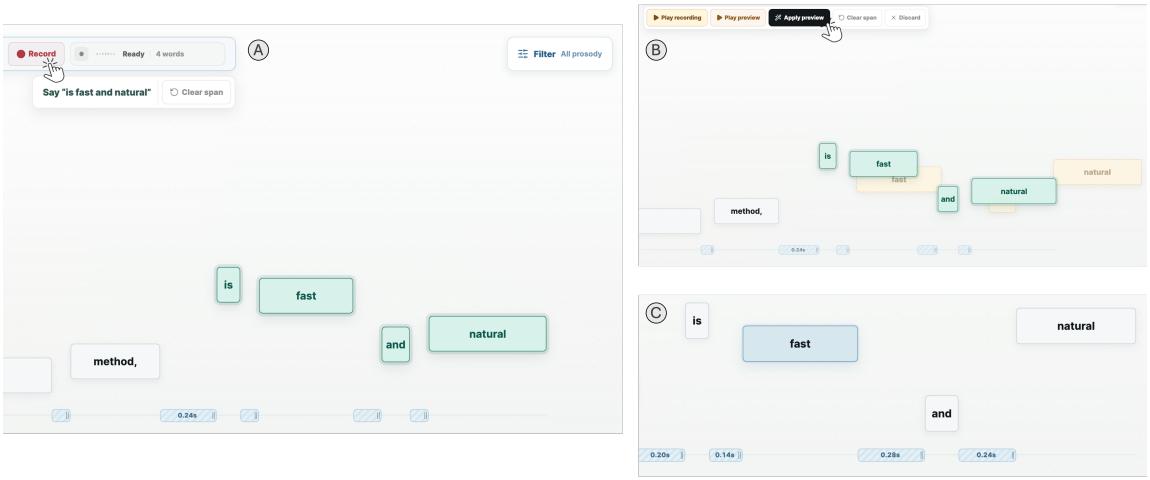


Fig. 3. The vocal demonstration workflow. (A) Selecting and recording target words: The user selects a span of words (e.g., "is fast and natural") and records themselves to convey target prosody. (B) Previewing and alignment: The recorded words are shown (in yellow blocks) overlaid on the target words (in green blocks), allowing users to audition the demonstration and synthesized preview before clicking "Apply preview". (C) Applied prosodic updates: Committing the edit updates the spatial canvas with newly aligned pitch heights (e.g., increased the pitch of "natural"), word durations, emphasis level, and updated pause indicators along the timeline.

preferred version. Alex next wants "However" to mark the transition from these benefits to the limitations of speech. He double-clicks the word to increase its emphasis and drags its right edge to lengthen it slightly (Figure 4A).

To explain the limitations more deliberately, Alex records the complete sentence. After comparing the recording with the current narration, he selects the second half of the sentence in the recorded overlay and applies it. He likes the delivery but wants more separation between the "linear, disorganized, and full of disfluencies," so he lengthens the gaps after "linear" and "disorganized" by dragging their pause regions rather than recording again (Figure 4B). The final sentence explains why transcripts are difficult to use. Its ending rises, but Alex wants a firm, falling ending when he introduces this limitation. He moves the word blocks for "make sense of" to lower their pitch levels (Figure 4C). Finally, he plays the complete paragraph to check that the energetic opening and the calm explanation of limitations could work together to enhance his speech.

4.2 Key Features

4.2.1 Speech Analysis and Visual Representation. ProsoLab represents speech as word blocks whose appearance encodes their delivery. Each block is linked to its position in the script, connecting acoustic measurements to the words users select and edit. The same waveform analysis procedure is applied to the initially generated speech, user recordings, and synthesized previews.

We first align the known script with the audio using forced alignment, which estimates when each word begins and ends (Figure 5a). These boundaries determine word duration and the pause between two words. We then extract fundamental frequency (F_0) from voiced frames using Praat and calculate frame-level root-mean-square energy in decibels. Pitch is converted to semitones relative to the median F_0 of the analyzed sentence. The median pitch of each word provides its overall pitch level, while five samples within its voiced interval form the word's internal pitch shape. For word-level emphasis, we estimate a weighted score for each word from its energy, pitch, and duration, all normalized

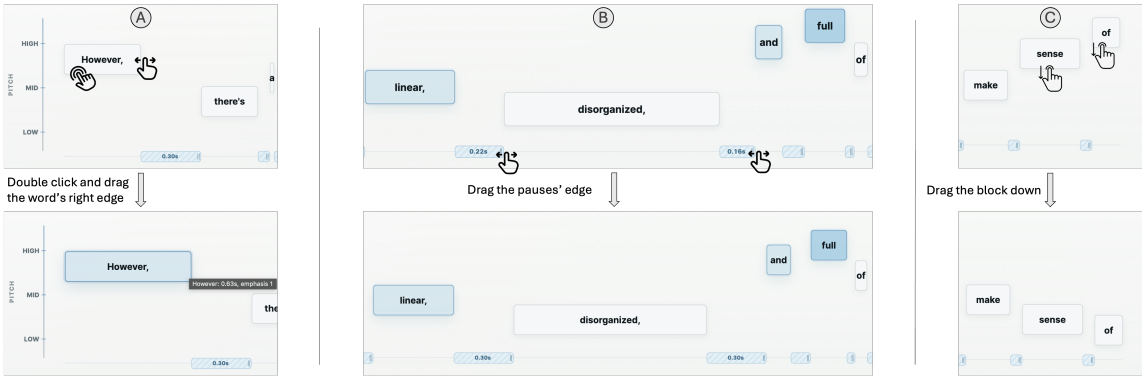


Fig. 4. Direct manipulation controls on the word blocks. Top rows show user interactions, and bottom rows show the resulting prosodic adjustments: (A) Word duration and emphasis: Dragging the right edge of a word block horizontally lengthens its duration, while double-clicking updates its emphasis level (indicated by fill color); (B) Inter-word pause: Dragging gap handles on the timeline adjusts silence duration between words; and (C) Pitch level: Dragging word blocks vertically adjusts their relative pitch.

within the sentence using the sentence's median and median absolute deviation. The score assigns energy a weight of 0.5, pitch 0.3, and duration 0.2.

On the canvas, a word block's width represents its duration, and the horizontal gap after it represents the following pause. Both use the same time scale. Vertical position represents relative pitch, with higher-pitched words placed higher. Darker color indicates higher emphasis. The emphasis is rounded and is limited to four display levels from 0 to 3. These mappings place speech measurements directly on the script rather than requiring users to locate words in a waveform. For recording-based and natural-language edits, yellow overlays show the candidate delivery compared to the current version. Corresponding words retain their script text, while their positions and widths may change during delivery.

4.2.2 Vocal Demonstration. The vocal demonstration lets users specify their desired delivery by speaking the corresponding words (Figure 5b). Commercial tools already support recording-guided delivery, including ElevenLabs Actor Mode and Murf Say It My Way [10, 25]. In the public documentation we reviewed, neither tool provides source code or algorithmic explanations that are sufficient to reproduce its transfer process. We implement an analysis-and-resynthesis pipeline that makes the transfer scope selectable and connects demonstrated changes to the controls used by the other editing methods.

Users can decide what to transfer either before or after recording. In the select-first workflow, the user selects a continuous word span and records only those words. In the record-first workflow, the user records the complete sentence and then selects the word areas to transfer. Selecting the complete sentence transfers all available changes. Changing the selection scope filters the changes from the recording, allowing the user to preview different scopes without recording again. In both workflows, the selection identifies where the transfer applies, while the recording supplies the delivery within that scope.

The transfer algorithm compares aligned word-level measurements from the current speech and the recording. Word timing is expressed as recorded-to-source duration ratios, while pause changes are represented as differences in duration. Pitch transfer separates movement across words from movement within a word. The relative pitch difference between corresponding words describes the pitch variation in a phrase. Within each word, five samples at normalized time positions describe its pitch contour. We center each contour on its own average value, then compare the recording

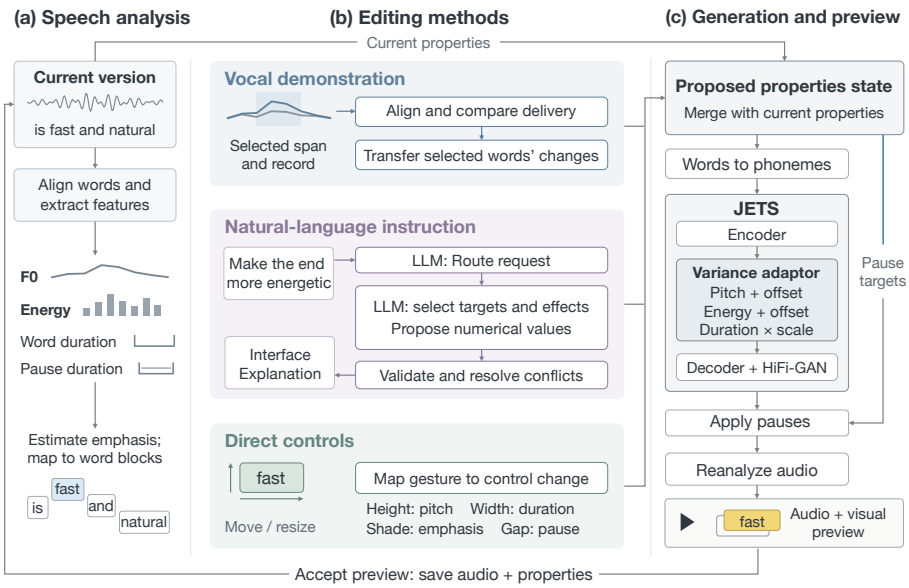


Fig. 5. The system architecture and processing pipeline. (a) Speech analysis: The current speech audio and transcript are aligned to extract acoustic features (F_0, energy, word duration, and pause duration), estimating emphasis levels to construct initial word blocks. (b) Editing methods: Vocal demonstration compares recorded audio to the original audio and transfers prosody to selected words; Natural-language instruction uses an LLM pipeline to propose numerical acoustic offsets; and Direct controls directly map gestures to acoustic variables. (c) Generation and preview: Proposed edits merge with original properties and drive a JETS acoustic model to synthesize speech and update the audio-visual preview.

to the source to distinguish a word's internal pitch variations from its overall pitch level. This transfers relative pitch movement without copying the user's absolute vocal range. For emphasis transfer, we estimate a prominence score from normalized pitch and loudness measurements, with syllable duration providing an additional cue. Each score is normalized separately within the source and recording, so it depends on how a word stands out within its speech track. These controls use the generation process described in Section. The resulting audio becomes the preview for user review.

Single-word pronunciation correction uses a separate recording path because prosodic variations cannot replace incorrect vowels or consonants. The user selects a word and records its intended pronunciation, and ProsoLab provides two candidates. The first transfers the recorded within-stress pattern while preserving the word's vowels and consonants. It compares aligned vowel regions in the original and recorded speech, updates the phoneme stress markers, and applies changes to vowel-level pitch, energy, and duration. For example, a recording that stresses the second syllable of "permit" can guide a change from *PER-mit* to *per-MIT*. The second supports changes to the phoneme sequence itself. A phoneme recognizer scores possible pronunciations based on the recording. These possible pronunciations come from dictionary entries and grapheme-to-phoneme conversions, supplemented by restricted phoneme variants supported by the JETS vocabulary. For example, for the word "externalization," this method combines the known stem "external" with two pronunciation variants of "-ization" and scores them against two saved recordings. Both candidates use JETS to synthesize the edited word using the synthetic voice.

4.2.3 Natural-Language Instruction. Natural-language instruction allows users to describe a desired delivery without having to set each control directly (Figure 5b). Users select one or more sentences or the whole paragraph, and enter an instruction. A large language model (LLM) first routes the request using the selected sentences' text, indexed words, and current sentence gaps. The router separates compound requests into operations and assigns each operation a sentence scope. For example, "Make it slightly faster and sound more energetic" is split between a pacing request and an expressive request.

Each operation is processed through a planning prompt that determines its detailed targets and numerical controls. Pacing proposals use speed ratios relative to the current delivery, with 1.0 indicating no change. Pause proposals specify target durations in seconds at word or sentence boundaries. For expressive requests, the planner identifies focus words and declares which parameters it will change. Its prompt guides local emphasis toward pitch and energy adjustments on those words, while requests for rising or falling intonation at the end of a sentence primarily affect the tone near the end. The planners propose magnitudes using predefined reference values and examples supplied in their prompts. For instance, the style prompt pairs "make the finding the main point" with approximate increments of +0.65 pitch and +1.5 energy on the relevant words. The prompt also provides a numeric range for each parameter. The planners do not receive current word-level acoustic measurements.

Deterministic backend rules validate and combine the proposals. Word indices and sentence scopes are checked, and numerical values are clipped to allowed ranges. For expressive edits, parameter values outside the declared set are discarded. Local energy changes are restricted to focus words, while neighboring support words may receive pitch or timing changes. For proposed changes that affect the same parameters and targets, the backend also ranks their scope specificity and instruction explicitness to decide whether to merge them or keep one.

The resolved plan forms an intermediate representation between the instruction and the speech output. It stores the speed ratio, pitch and energy increments of each word, as well as the specific pause location and duration. The interface directly retrieves the read-only description from the plan and links it to the affected words (Figure 2). The plan then enters the generation process described in Section. The descriptions show the proposed changes, while the yellow comparison overlay shows measurements from the generated audio. Users can view the explanations and audio results sentence by sentence, and accept or reject the changes.

4.2.4 Direct Controls. Direct controls allow users to refine speech by editing word blocks and the gaps between them on the canvas. Each gesture specifies both its target and an adjustment to one control (Figure 5b).

Double-clicking a word cycles through four emphasis levels, numbered 0–3 and represented by increasingly darker shading. Previously, we mapped continuous emphasis scores to discrete levels; this step is the reverse. Dragging a word upward or downward changes its overall pitch control in proportion to the vertical displacement, while preserving any stored within-word pitch contour. Dragging the right edge of a word changes its duration. The ratio of this duration to the current version's duration scales the stored duration control. We set upper and lower limits for the duration control of each word based on the phoneme information of the word. Dragging the pause region between adjacent words sets the silence duration at that boundary.

The canvas updates during each gesture and keeps the requested positions and sizes during preview and after acceptance. Even if the generated speech after editing cannot reproduce the exact requested parameters, this setting keeps the canvas stable while users compare it to the audio preview.

4.2.5 Speech Generation and Preview. As shown in Figure 5c, the three editing methods update a persistent control state that ProsoLab uses to generate speech with a pretrained JETS model [22]. Each word stores pitch and energy differences

together with a duration multiplier. Pause targets are stored at word boundaries, while demonstrated within-word pitch contours and pronunciation corrections are stored separately.

To synthesize an affected sentence, we convert its text into phonemes and map each word to its corresponding phoneme range. The word's pitch and energy controls are added to JETS's predictions for those phonemes in the model's normalized feature spaces. Changes measured in semitones or decibels are converted using the checkpoint's normalization statistics. Stored within-word pitch differences are interpolated across the word's phonemes and added to its overall pitch control. Duration multipliers scale the predicted phoneme durations. When lengthening a word, we allocate more additional time to vowels than to consonants, and the limitations of phoneme categories restrict the upper limit of word extension. Shortening applies a uniform multiplier within the word.

Pause editing operates on the generated waveform. We find relevant word boundaries and insert or remove silences to approximate the stored pause target. Short fades smooth the resulting boundaries. At the paragraph level, the affected sentence is merged with the current versions of the remaining sentences and their pauses.

The edited audio is reanalyzed to obtain playback timing and acoustic properties. The preview remains separate from the current version while the user compares them. Accepting a preview saves the reviewed audio together with its associated control state. Subsequent edits start from this accepted version, allowing users to edit iteratively without having to rebuild their previous adjustments.

4.3 Implementation Details

ProsoLab is a web application with a React frontend. Its word-level editing canvas is built with React Flow, and Vite is used for development and production builds. The frontend communicates with a Python HTTP backend that runs speech inference using PyTorch and a modified ESPnet implementation of JETS [20]. We use the pretrained imdanboy/jets [1] checkpoint to generate audio at 22.05 kHz. Recorded demonstrations are aligned using TorchAudio's MMS forced aligner [2], and pitch is extracted using Praat through Parselmouth [3]. Pronunciation analysis uses a wav2vec 2.0 model through Hugging Face Transformers [4]. Pronunciation candidate generation combines CMUdict entries with grapheme-to-phoneme conversion and a set of phoneme alternatives. The natural-language planner is configured to use gpt-5.6-luna via the OpenAI API [5], with a medium reasoning effort. Its JSON responses are validated by the backend before being translated into speech controls.

5 User Evaluation

We designed a two-part within-subjects study to examine how the three editing methods support speech-editing goals and how users work with them in an integrated interface. The first part compares methods on assigned sentence-level tasks. The second allows participants to choose and combine methods while editing a paragraph. The study addresses three research questions.

RQ1. How do vocal demonstration, natural-language instruction, and direct manipulation differ in supporting speech-editing tasks with different prominence or timing goals? We examine perceived goal achievement and control, together with editing efficiency, iteration count, and workload, and consider how task characteristics relate to method suitability.

[1] https://github.com/imdanboy/jets
[2] https://docs.pytorch.org/audio/2.9.0/tutorials/forced_alignment_for_multilingual_data_tutorial.html
[3] https://parselmouth.readthedocs.io/en/stable/
[4] https://huggingface.co/docs/transformers/en/model_doc/wav2vec2
[5] https://developers.openai.com/api/docs/models/gpt-5.6-luna

RQ2. How do users choose, switch between, and combine the three editing methods during open-ended speech editing? We examine initial method choice in relation to editing goals, the events that prompt switching, and the roles methods play at different stages of editing.

RQ3. How do users perceive the usability, learnability, transparency, and overall usefulness of our interface integrating the three methods? We also examine how users interpret the visual and audio feedback when evaluating edits.

5.1 Participants and Study Setup

We recruited 12 participants (6 males, 6 females) from the university's social media. They are all graduate students with diverse academic backgrounds, including computer science, linguistics, art, and design. All participants have created spoken content for practical use and have previous experience with TTS tools (detailed in Table 4). Each participant received the equivalent of 20 USD as compensation. Half of the study took place in person in a quiet meeting room, and the other half took place online while participants were in quiet places. They all share the same link of ProsoLab and open it on their own computer. During the study, their screens and sound were recorded using meeting software. Each part began with informed consent, an introduction to the study process, and the background questionnaire. Before Part 1, they had a tutorial about the prototype and practiced for about 10 minutes. The whole study has been reviewed and approved by the IRB at our institution.

5.2 Part 1: Comparing Editing Methods

Each participant will complete three method blocks, using vocal demonstration, natural-language instruction, and direct manipulation in separate blocks. Each block will contain one task set containing one practice task followed by two comparison tasks. This gives each participant three practice tasks and six comparison tasks. Participants will use only the assigned editing method within each block.

We prepared three task sets. Each contains one comparison task focusing on prominence and one focusing on timing. Each task combines a localized delivery goal with a broader expressive goal. For example, a prominence task asks participants to make a selected phrase stand out while conveying confidence, whereas a timing task asks for a local change in pacing within a particular overall rhythm. The task specifies the required delivery method but not the numerical parameter requirement. In addition to the task description, each task includes a speech to edit and its script. Participants need to listen to the speech and edit it using the assigned method, previewing and revising it until they decide to stop. After each task, participants will complete a questionnaire that includes 5 seven-point rating items to assess whether changes remained within the intended scope, the naturalness of the result, the controllability of the change, and the suitability of the method for the task. Participants also reported their reasons for stopping. After each method block, they will complete the NASA-TLX to report workload [14].

The study used Latin-square counterbalancing for method order and task-set assignment (Appendix). The practice task will always come first. Participants P1 to P6 will complete the remaining tasks in their listed order, while P7 to P12 will complete them in reverse order. Approximately 45 minutes are allocated to the tasks, followed by a five-minute interview about which goals were easier or harder to achieve with each method and whether any goals were poorly supported by all three methods.

5.3 Part 2: Open-Ended Editing

Participants then had approximately 15 minutes to edit a paragraph with all three methods available. They chose from the three provided scripts based on their interests. Unlike Part 1, this part did not provide a specific task. Participants needed to listen to the speech, use the thinking-aloud protocol to identify delivery problems they consider worth addressing, and describe their intended changes before choosing a method. They may continue with one method or switch and combine methods as their editing progresses.

After editing, participants completed a questionnaire that included two UMUX-LITE items assessing whether the system meets users' requirements and is easy to use [21]. Four additional seven-point items assess whether an edit matches the intended result, perceived control, knowing which method to use, and alignment between the canvas and the audio. Participants also reported which method contributed most to their final result. A 15-minute interview reviewed specific editing moments to discuss initial method choices and the reasons for switching to or continuing to use a particular method. It also included whether hearing an intermediate result changed the participant's goal, how they decided when to stop, and what they felt was missing in the current prototype. The full task scripts, questionnaires, and interview questions are in the Appendix.

5.4 Data Analysis

System Log. ProsoLab recorded timestamped editing actions, previews, and accepted versions, together with affected word positions. We reconstructed task-level behavior in Part 1 and sentence-level editing sequences in Part 2. Method switches counted changes between editing attempts within the same sentence, including revisits. Unsupported instruction requests and pronunciation-panel openings without recording were excluded. Paragraph-wide edits reset the affected sentence sequences.

Word-level Editing Measures. For each editing event e, let W_e denote its affected word positions. *WEC* measures cumulative word-level editing count:

$$\mathrm{WEC} = \sum_{e} |W_e|. \tag{1}$$

UEW measures the number of distinct word positions edited at least once:

$$\mathrm{UEW} = \left| \bigcup_{e} W_e \right|. \tag{2}$$

We further define *RI* as:

$$\mathrm{RI} = \frac{\mathrm{WEC}}{\mathrm{UEW}}, \tag{3}$$

represents the average number of operations per edited word position, including the first edit. Pause edits were excluded from these word-position measures because they occur at word boundaries.

Statistical Analysis. Participants were the unit of inference. In Part 1, we excluded the practice task and aggregated the two comparison tasks within participant and method for across-method comparisons. Three-method comparisons used Friedman tests, with exact permutation tests for word-level measures. Two-sided exact Wilcoxon signed-rank tests examined pairwise differences and prominence–timing differences within direct manipulation. We report Kendall's W and paired rank-biserial correlations as effect sizes and applied Holm correction within related test families. Part 2 inferential comparisons used participant-level method proportions. Interface questionnaire results were summarized descriptively, and additional word-level and sequence analyses were treated as exploratory.

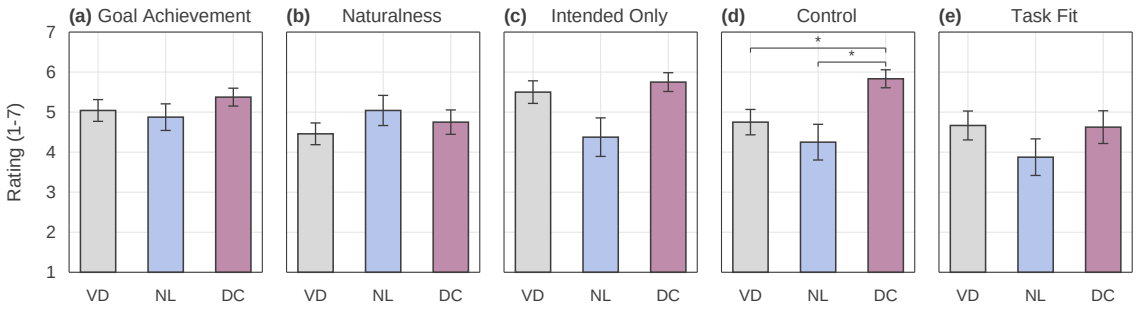


Fig. 6. Subjective outcomes in Part 1. DC received higher perceived-control ratings than both other methods. The remaining four outcomes showed no significant method differences. Brackets mark pairwise differences with Holm-adjusted $p < .05$; detailed tests appear in Tables 2

Qualitative analysis. Two authors reviewed the full interview transcripts. We combined research-question-guided coding with codes developed from participants' accounts [12]. Meaningful passages were the coding unit and could receive multiple codes. The codes across participants were compared and categorized into themes, retaining descriptions that support or challenge the interpretations being formed. For Part 2, interview and think-aloud accounts were related to logged sequences to examine method choices and subsequent editing decisions. Themes were interpreted alongside the quantitative findings to answer research questions.

6 Results

Across the study, participants used three editing methods: Direct Controls (DC), Vocal Demonstration (VD), and Natural-Language Instruction (NL).

6.1 RQ1: How Did the Three Methods Compare in Assigned Tasks?

6.1.1 *Quantitative Comparisons of Outcomes and Editing Activity.*

Perceived outcomes and workload. DC received higher perceived-control ratings than both VD and NL. Mean ratings on the seven-point scale were 5.83, 4.75, and 4.25, respectively. The comparisons with VD and NL were significant after Holm correction ($p_{\text{Holm}} = .029$ and .023, respectively; Figure 6). We found no significant method differences in goal achievement, naturalness, intended-only changes, or method–task fit (Figure 6). NASA-TLX scores also showed no significant method difference ($\chi^2(2) = 1.17$, $p = .558$).

Editing time and recorded iterations. NL took longer than both other methods, averaging 249 seconds compared with 148 seconds for VD and 146 seconds for DC. After subtracting identifiable backend latency, NL still took significantly longer than DC (198 versus 118 seconds), but its difference from VD (136 seconds) was not significant. Recorded iterations followed a different order: DC had the highest mean (7.96), followed by NL (4.17) and VD (1.71), with significant differences between all three pairs (Figure 7). These counts represent method-specific interaction cycles rather than equivalent units of effort.

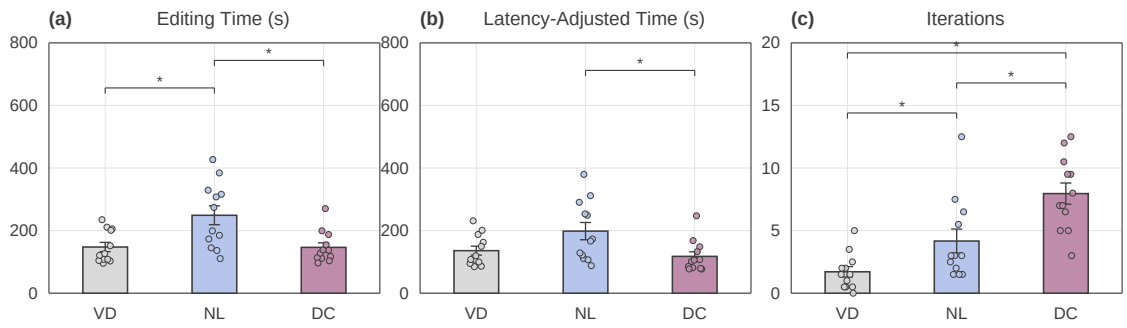


Fig. 7. Editing processes in Part 1. NL took longer than both other methods in total elapsed time (a), and longer than DC after identifiable backend latency was subtracted (b). DC involved the most recorded iterations (c). Bars show means across participants, error bars show standard errors, and points show individual participants. Brackets mark pairwise differences with Holm-adjusted $p < .05$. Iterations represent method-specific interaction cycles.

Word-level editing coverage. We measured unique edited words (UEW) as the number of distinct word positions involved in recorded edits, counting each position only once, even when it was edited repeatedly (Figure 8). Pauses were analyzed separately. Median counts were 34 for VD, 20.5 for NL, and 10.5 for DC. Both VD and NL involved more unique word positions than DC ($p_{\text{Holm}} = .009$ and .031, respectively). Across-method differences in WEC and re-edit intensity were not significant after correction.

Timing and prominence tasks within direct controls. Within DC, prominence tasks involved more cumulative word-edit operations than timing tasks ($Mdn = 16$ versus 10.5, $p_{\text{Holm}} = .046$) and higher re-edit intensity ($Mdn = 3.27$ versus 1.87, $p_{\text{Holm}} = .015$) (Figure 9a–c). UEW counts did not differ significantly between task types ($Mdn = 5$ versus 5.5). The parameter-level analysis showed more pitch and emphasis operations in prominence tasks (Figure 9d; both $p_{\text{Holm}} = .012$), whereas duration operations showed no significant difference. The separate repeated-adjustment analysis also found increases in both pitch and emphasis: median repeated pitch adjustments increased from 0 to 2.5 ($p_{\text{Holm}} = .027$), and repeated emphasis adjustments increased from 0 to 3.5 ($p_{\text{Holm}} = .023$). These comparisons concern task differences within DC, not differences in task sensitivity between methods.

Summary of Quantitative Comparisons of Three Methods. The three methods differed in both perceived control and editing behavior. DC produced significantly higher perceived-control ratings than the other two methods, while the remaining task-level subjective outcomes and workload did not differ significantly. NL showed the longest editing times, whereas DC involved the largest number of explicit interaction cycles. At the word level, VD involved significantly broader lexical coverage than DC. Within DC, prominence tasks produced significantly higher cumulative editing and re-edit intensity, while the number of unique edited word positions did not differ significantly. These additional operations were concentrated particularly in pitch and emphasis adjustments.

6.1.2 *Qualitative Results of Method Suitability.*

Specifying local timing and pitch changes. P6 and P11 described DC as useful when they could identify a specific adjustment. P6 found visible values helpful for following fine changes, particularly pause durations. P11 described directly setting a pause duration or raising a word's pitch as convenient when the desired adjustment was already

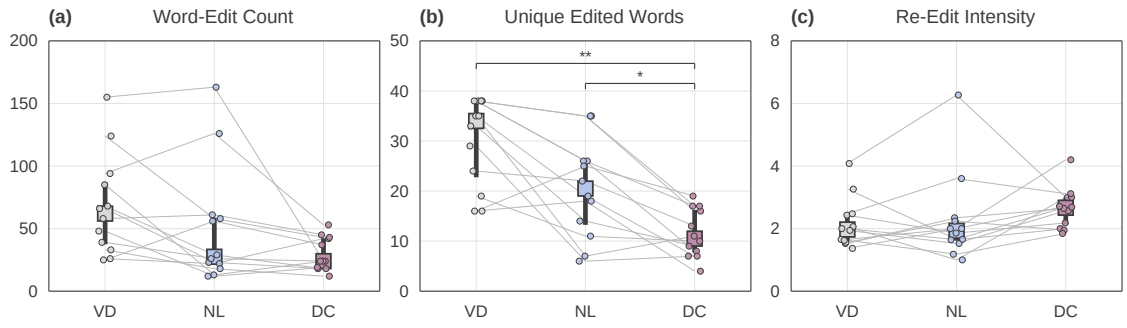


Fig. 8. Word-level editing in Part 1 for 12 participants with complete word-position records. Both VD and NL involved more unique edited word positions than DC (b). Omnibus method effects on WEC (a) and re-edit intensity (c) were not significant after Holm correction. Points and connecting lines show repeated measures from individual participants; larger markers show medians and intervals show interquartile ranges. Brackets mark Holm-adjusted pairwise differences ($^{*}p_{\text{Holm}} < .05$, $^{**}p_{\text{Holm}} < .01$).

clear. In contrast, P11 found it cumbersome to repeatedly describe pause adjustments through NL when the duration still needed testing. P5 similarly described having to "first turn it into parameters, and then express those parameters in language." Direct adjustment also required auditory checks to build a mental model between the visual preview and the intended delivery. P8 reported uncertainty about which parameter would produce the desired delivery and listened after each change. P12 described an unexpectedly large effect from a small duration adjustment: "My subjective impression was that I had shortened it only a little, but it immediately sounded very fast."

Expressing delivery without selecting parameter values. P1 and P7 described VD as useful for expressing rhythm or for smooth delivery without having to decide on individual parameter settings. P7 explained, "I was not sure whether it should be faster or slower, so I read it smoothly according to my own understanding and let it imitate me." P11 also found VD useful for the intonation of one or two words when the corresponding dragging operation was unclear. However, P11 found whole-sentence demonstration difficult when unsure how to perform the requested delivery, and P3 described re-recording after reading errors or hesitation. NL offered another way to work with goals when they are uncertain about the effect, but want to give it a try and establish the overall style. P7 explained: "I could give it a general direction and let it decide. I would just judge whether it was good and whether to accept it." P11 likewise considered descriptions of relaxed or tense delivery suitable for NL, even if they cannot express exactly what the request is about.

Preserving continuity during local edits. Participants described difficulties changing selected words while keeping the surrounding speech natural. With DC, P5 described adjusting surrounding words to establish the intended contrast: "Sometimes, after making the change, you find that the contrast with the surrounding context has not increased, so you need to change the surrounding context." P9 and P10 also reported that local emphasis could sound disconnected from the rest of the sentence. P10 found that individual adjustments could produce the desired emphasis, yet the sentence sounded abrupt because the surrounding delivery did not fit those changes. With NL, P10 considered the overall delivery produced through NL more coherent, but reported little emphasis on three explicitly requested words, even when the sentence shifted toward the requested style. With VD, P7 described difficulty choosing how much speech to record when editing words within a long sentence. Reading the entire sentence could disrupt its fluency, while demonstrating only the selected words could yield an unsatisfactory result in context.

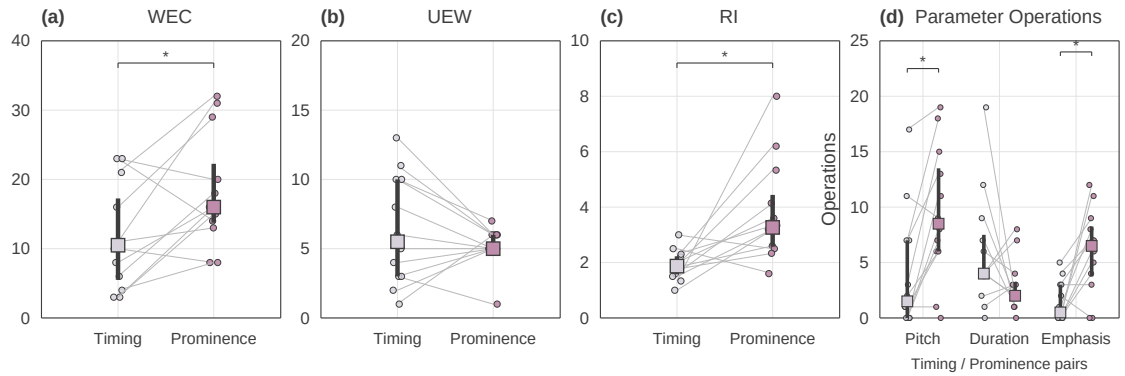


Fig. 9. DC in timing and prominence tasks. Prominence tasks involved more cumulative word-edit operations (a) and greater re-edit intensity (c), with no significant difference in UEW (b). Pitch and emphasis operation counts were also higher in prominence tasks (d). Points and connecting lines show repeated measures from individual participants; larger markers show medians, and intervals show interquartile ranges. Brackets mark differences with Holm-adjusted $p < .05$. Panel (d) shows operation counts.

6.2 RQ2: How Did Participants Choose and Combine Methods During Open-Ended Editing?

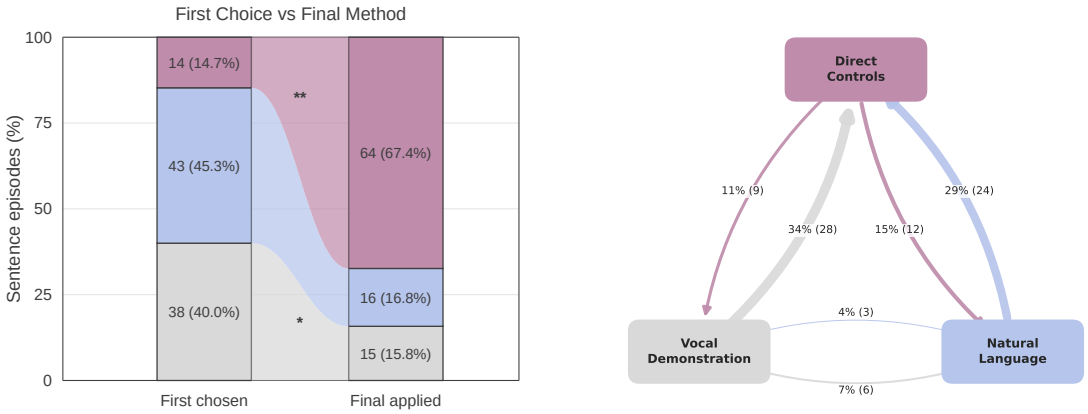
During open-ended editing, the 12 participants edited 95 sentences in total. Among 51 natural-language requests with supported responses, the planner classified 39 (76.5%) as style-oriented. Six requests (11.8%) combined pace and style, and six (11.8%) concerned pace alone. In terms of the scope, 7 attempts (13.7%) addressed multiple sentences or a paragraph, 39 (76.5%) addressed a whole sentence, and only 5 (9.8%) directly targeted a local word or phrase. Of the 42 accepted VD edits, 26 (61.9%) applied delivery from recordings to selected regions and 11 (26.2%) used continuous-span imitation. Five (11.9%) were pronunciation repairs. These distributions describe use throughout the session. The following analysis examines how participants selected and combined methods during editing.

6.2.1 Initial Method Choice Reflected Participants' Confidence in Expression. We examined the first method used to edit each sentence. As shown in Figure 10a, across 95 edited sentences, NL was used first for 43 sentences, VD was used first for 38 sentences, and DC for 14.

Participants began with NL when they could describe the intended style but were unsure how to produce it through speech or DC. P7 began editing the concluding sentence this way, explaining: "I know roughly what style is needed, but I do not know how to adjust it specifically, so I will type first." P10 also began with this method to establish a fairy-tale atmosphere throughout the paragraph.

Participants began with the VD when they believed they could achieve the intended delivery on their own, or when it was hard to describe the intended effect. P11 chose recording for the opening question because the sentence was short and could be spoken directly. P7 began editing "Be careful" with a recording because the intended delivery was difficult to describe.

DC could be the starting method when the existing speech already met part of the participant's goal and they had a clear idea of the specific adjustments needed. P1 judged the overall rhythm of one sentence acceptable but wanted to lengthen a particular part, so they began with manual adjustments to timing and emphasis. P7 identified insufficient emphasis on "have a chance" and chose to adjust it on the canvas first. In the interview, P9 described DC as the most direct editing method, consistent with its use at the beginning of seven sentences.



(a) First-choice and final-applied editing methods.

(b) Adjacent transitions between editing methods.

Fig. 10. Method progression and switching during open-ended editing. (a) Comparison of first-choice and final-applied editing methods across sentence episodes. Asterisks indicate significant participant-level differences between first-choice and final-applied proportions ($^{*}p_{\text{Holm}} < .05$, $^{**}p_{\text{Holm}} < .01$; two-sided Wilcoxon signed-rank tests with Holm correction). (b) Adjacent transitions between the three editing methods. Arrow labels show the percentage and count of all observed method switches for each transition direction.

6.2.2 Methods Switching Supported Repair, Exploration, and Newly Identified Subgoals. As shown in Figure 11, participants made 82 method switches within sentences during the main editing task. DC was the destination of 52 switches (28 from VD and 24 from NL), VD was the destination of 12 switches (9 from DC and 3 from NL), and NL was the destination of 18 switches (12 from DC and 6 from VD). Participants made 8 switches between global and local edits.

Repairing results and overcoming editing difficulties. Participants switched methods when the current approach did not produce the delivery they wanted. P7 rejected a NL to make the sentence sound more convincing, saying, "It feels very wrong, so I reject it." P7 then performed the sentence through VD and accepted the result as better than the previous versions. P10 likewise discarded a recording-based result because it differed from the performed example, then adjusted the sentence manually. In both cases, the editing goal remained the same, while the participant changed how they pursued it.

Repair also followed problems that became apparent after an edit. P9 explained that "after adjusting the intonation (manually), the whole sentence was actually less fluent." The participant then used NL to address the poor word-to-word connection. P11 encountered a different problem after requesting faster speech by NL: some words seemed too short to sound complete. P11 responded by lengthening them through DC.

Participants could also switch methods when they knew what they wanted to hear but did not know how to produce it with the current method. While adjusting "we know" via DC, P7 said, "I really do not know how to adjust it," and turned to recording. Similarly, P8 explained that adjusting words individually was troublesome and that they did not know how to make the desired changes, so they read the sentence as intended instead. P11 encountered a similar difficulty when trying to perform an angry tone, explaining, "I cannot express the tone particularly accurately," before

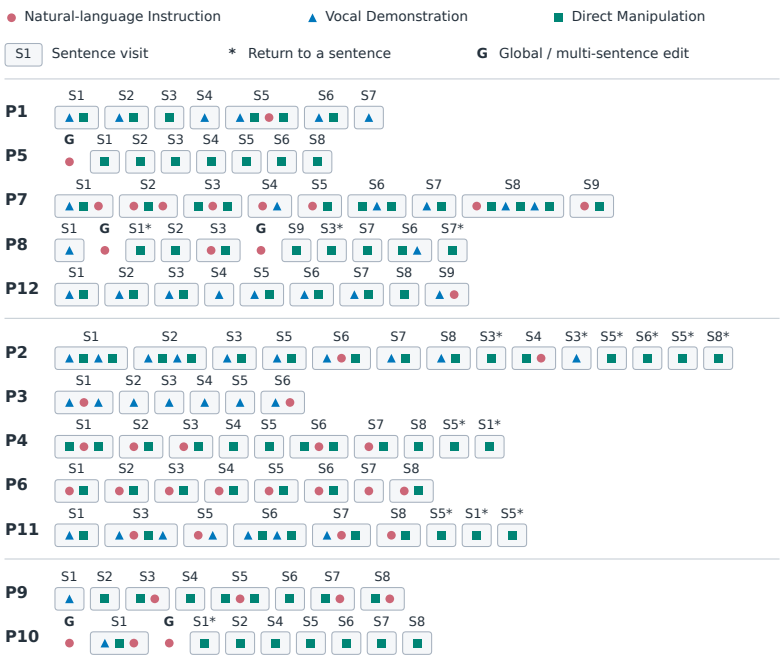


Fig. 11. Method-use sequences during open-ended editing across 12 participants. Horizontal dividers separate three groups of participants who edited different paragraphs. Sentence labels refer to the same text within each group. Each row shows one participant's editing sequence from left to right. Each symbol groups consecutive actions that use the same method and scope, including unaccepted attempts. Spacing represents order rather than duration.

using NL. In these situations, switching methods provides a way to continue working when it becomes difficult or uncertain to continue using the current method.

Exploring alternatives and developing preferences. A switch could also reflect interest in another possibility. After manually editing one sentence, P7 said: "I actually think this sentence is almost there, but perhaps there is a better solution." P7 then requested a more exciting delivery through NL. However, after listening to the generated result, P7 rejected it and returned to manual editing, as the previous version was better. The first switch explored an alternative to an approximately satisfactory result. The return to manual editing followed rejection of that alternative.

Another result could also provide a reference for subsequent editing. After a natural-language edit, P1 identified one occurrence of "know" as sounding good and used it as a reference when manually adjusting the other occurrence. Similarly, P7 stated that "I used natural language and suddenly found that it adjusted the speech better than I did. Then I would imitate its modifications to modify others." Here, the result helped the participant decide how to continue editing another word.

The exploration can also be related to visual patterns. P5 began with NL to make the paragraph sound "strange" and then used DC to arrange word-level patterns. In the interview, P5 explained that the generated speech was already strange enough, but that they wanted to explore which visual arrangements produced that effect. The manual

work included descending patterns and alternating emphasis. Here, moving to another method further supported the investigation of the output rather than the correction toward better delivery.

Developing subgoals and choosing how much to change. A better result could reveal a more specific next step. After recording "Be careful, and make sure you look out for each other," P7 judged the result better than before, then identified "make sure" as needing stronger emphasis and moved to DC. The switch did not mean rejection of the preceding result. Similarly, P8 initially wanted an angry delivery via NL. Listening to the results led them to identify "friends" and "family" as words that needed to be separated and emphasized.

Participants also considered how many changes were still needed and what disruptions another attempt might cause. When refining "safely," P7 explained that another recording would make other changes that would need to be manually deselected, and chose DC instead. P1 similarly explained that another recording was unnecessary because only a small adjustment remained.

A next step could also extend beyond the initial local target. P11 first recorded "fifteen minutes" to emphasize that phrase, then judged the whole sentence too slow for the intended urgency and used NL to request a faster delivery. P9 similarly moved from a manual timing adjustment to an instruction emphasizing several verbs and natural delivery throughout the sentence.

6.2.3 Direct Controls Were Often Used for Final Fine-Tuning. The final applied method was the last method whose output was accepted for a sentence. Of the 95 edited sentences, direct control was the final method for 64 (67.4%), NL for 16 (16.8%), and VD for 15 (15.8%) (Figure 10b). Two-sided Wilcoxon signed-rank tests confirmed a significant increase in the proportion of DC from first choice to final application ($p_{\text{Holm}} = .0059$), paired with a significant decrease in VD ($p_{\text{Holm}} = .0313$); the reduction in NL was not significant ($p_{\text{Holm}} = .0811$). Rather than initiating edits, the DC functioned primarily as a terminal refinement layer following broader initial passes initiated via voice or text.

DC could complete an edit by addressing a remaining local requirement after a broader change. P7 retains better NL results than the original version, but further adjustments were needed so they used DC to refine the sentence. P10 decided to complete the NL edits before making fine adjustments because they said broad NL changes affected previously edited details. Therefore, the final use of DC reflected both remaining local work and an effort to avoid disturbing those adjustments with another broad edit.

An edit could also end with NL or VD when the result met the participant's current requirements. P12 used NL to slow the final sentence and then described the result positively. For another sentence, P12 initially planned to manually refine the recording result. However, after a few trials, P12 judged that the result was not better than before, "the benchmark was the voice I originally recorded," so they rejected the manual edit and kept the recorded one.

Stopping did not always indicate that every problem had been resolved. P7 stated "There are two ways I judge this. One is that it sounds close to what I imagined, or even better. The other is that I really cannot make it any better, so I stop." P8 described ending adjustments when repeated changes seemed to make little difference or when further detailed work felt too costly. P10 similarly reported stopping when the overall effect was sufficiently close to the goal, and they did not know what else to change. Therefore, the final applied method alone does not establish complete satisfaction or a general preference for that method.

6.2.4 Participants Valued Methods Contributions Beyond the Most Frequently Used Method. When asked which method contributed most to their final editing results, four participants selected NL, four selected VD, and four selected DC. For each participant, we compared this response with the method that most often supplied their last confirmed sentence

application. The two differed for 7 of the 12 participants. P8 explicitly distinguished contribution (NL) from frequency of use (DC). They explained that NL established an overall direction for the delivery, stating, "Without an overall direction, it is difficult for me to make fine adjustments." For P8, the method's value lay in enabling subsequent refinement. P10 described manual and AI contributions as approximately equal, but he explained that part of the reason for choosing DC was the time spent adjusting individual words. These statements link perceived contributions to the method's role and to participants' engagement in using the method.

6.3 RQ3: How Did Participants Experience the Integrated Interface?

6.3.1 Participants Reported High Overall Usability and Clarity. The mean UMUX-LITE score was 86.11 (Mdn = 83.33, range 75–100). On the seven-point questionnaire items, participants reported high ratings for system capabilities (M = 6.00), ease of use (M = 6.33), agreement that edits matched the intended result (M = 6.00), perceived control (M = 5.92), clarity about which method to use (M = 6.33), and agreement between the canvas representation and the resulting audio (M = 6.08).

6.3.2 Practical Usefulness Varied with the Information to Convey. Participants identified uses for the interface in preparing spoken materials. P7 saw value in editing narration for paper videos, describing how correcting a few words in other tools could require regenerating speech and assembling the resulting clips. Local editing offered a way to work on those details within the speech. P3 anticipated a different use in preparing materials for experiments on sentence focus. They valued the possibility of producing more consistent examples than their own repeated recordings: "If I record with my own voice, each recording may be different."

The value of detailed editing depended on what the speech needed to communicate. P8 explained that a basic text-to-speech tool already served the main purpose of conveying information in demonstration videos. They saw additional value in using pauses and emphasis to guide viewers' attention, and a greater need for expressive control in dialogue: "When you need to convey emotion, I think it is very useful." P10 similarly anticipated using NL to make their speech faster and more confident.

Participants also identified needs that the available editing controls did not fully address. P7 wanted a wider choice of voices. P10 wanted finer control over the speech details that conveyed emotion, describing qualities that they could not achieve through the available pitch, duration, and emphasis adjustments.

7 Discussion

7.1 Speech Editing Should Support Clarifying Intended Delivery from Imperfect Input.

Our findings suggest that speech editing should support users who can judge whether a delivery suits their goal but cannot reliably explain or perform it. This distinction connects to Hutchins, Hollan, and Norman's account of directness as a relationship between users' intentions and the facilities an interface provides for achieving them [16]. In speech editing, a familiar input action can still require an unfamiliar translation. Natural language instruction requires converting an intended change into acoustic parameters before expressing it. Direct controls required participants to determine which adjustments would produce the desired sound. Vocal demonstration offered a way to convey coordinated changes without specifying their acoustic values, but required participants to produce those changes themselves. Therefore, providing several methods gives users different ways to express a goal while leaving open whether any available method matches what they can reliably communicate at that moment.

The vocal demonstration makes this problem particularly clear because the example can contain both intended expression and unintended features. Reproducing the demonstration accurately could preserve the feature that the participant did not want. A demonstration should therefore be treated as evidence from which the intended change can be clarified. Selecting the properties to transfer, as our interface allows, addresses part of this problem. Yet an unwanted hesitation can occur within the same timing pattern that the user wants to convey. Supporting such cases requires a method to extract only the parts the user wants to convey in the example, such as only the emphasis levels of a few words.

Future speech editors could make users' judgments a more explicit part of specifying changes. When a description leaves the delivery uncertain, the system could offer contrasting versions to help the user make a particular choice, such as where the emphasis should fall. For a demonstration, users could indicate which parts convey the intended expression and which should be revised. Subsequent proposals could build on those decisions, reducing the need to repeatedly describe or perform the entire change. The broader design challenge is to preserve the user's intended contribution while allowing the example itself to be imperfect. Evaluating these interfaces should therefore include how effectively users can achieve a desired result from incomplete descriptions or imperfect performance, and how much work they must do to distinguish intended features from unintended ones.

7.2 A Local Editing Goal Can Require Collateral Changes Beyond Its Target.

The location where a user wants an effect can differ from the region that needs adjustment to produce it. The findings suggest that making one word stand out can involve deciding how other parts of the sentence should contribute to that effect. Prosody research describes prominence through a word's strength relative to surrounding words [8]. Our findings identify a consequence of this relationship: precise control over where an operation acts does not necessarily confer precise control over its perceived effect. A participant may increase a selected word's emphasis yet still find that it does not stand out appropriately or fit the sentence's delivery. This distinction helps interpret the higher perceived control reported for direct manipulation alongside participants' remaining difficulties with the resulting speech. Therefore, precision in speech editing needs to account for the relationship the user wants to create between a target and its context.

Future interfaces could distinguish the focus of an editing goal from the scope of the changes proposed to achieve it. Selecting a phrase could identify where the user wants attention to fall, while a separate choice could allow adjustments to nearby speech. For example, a request to emphasize a phrase could produce a proposal that combines changes to the phrase with adjustments to adjacent pauses or surrounding emphasis. The interface could show how these supporting changes relate to the selected goal and provide a preview within the full sentence. This would extend the role of the word-aligned representation from exposing separate controls to helping users assess coordinated revisions.

Coordinated editing introduces a tension between achieving the local goal and preserving satisfactory speech. Leaving every surrounding word unchanged may limit the intended contrast, while changing the surroundings may disturb another aspect of the delivery that the user values. When these requirements conflict, the system could present alternatives, making the consequences available for comparison. Evaluation should examine whether a revision communicates the intended focus and maintains an acceptable surrounding delivery, alongside the effort required to achieve it.

7.3 Speech Models Should Build on Earlier Editing Decisions.

Our findings suggest two ways that earlier decisions could support continued editing. A model could preserve valued aspects of a delivery when responding to another request and use an accepted example to guide related changes elsewhere. Both capabilities could reduce the need for users to repeatedly establish an effect they have already achieved.

Recent speech-editing research already addresses localized instructions and preservation of unrequested regions or properties [40]. Our findings motivate examining how such capabilities support requirements that develop across successive revisions. A user may first establish emphasis on a phrase and later request faster delivery without wanting to lose that emphasis. The generation process would need access to the current speech and the earlier requirements that remain relevant. An interface or editing controller should automatically supply those requirements in the new request.

Accepted examples could also guide editing beyond the sentence in which they were produced. In ProsoLab, only the natural language instructions could be used across the paragraph. For vocal demonstration and direct controls, the reusable information should also concern how the speech communicates its purpose. Such assistance could help users establish a desired delivery using a few examples and focus subsequent work on passages that require different treatment. Future interfaces could let users identify which aspects of an example should guide later revisions and where to apply them.

7.4 Limitations

Our prototype uses JETS as its synthesis backend, which has high controllability but lacks the explicit emotional conditioning available in some recent TTS models. This constraint may have contributed to participants' difficulties in achieving certain emotional effects and influenced their judgments of the method's suitability. Evaluating the interface with more expressive synthesis models would help determine how these capabilities affect method use and editing outcomes.

The evaluation used provided English sentences and paragraphs in a laboratory setting. Participants could develop their own editing goals in the second session, but they did not bring their own scripts or complete an entire production project. Unfamiliarity with the supplied text could affect their ability to perform a demonstration, while the laboratory setting could influence how much effort they invested before stopping. Studies involving participants' own materials would help examine how the use of methods relates to practical requirements, such as fitting narration into a video or maintaining consistent delivery across a longer piece.

8 Conclusion

We presented ProsoLab, an integrated system for editing the expressive delivery of synthesized speech. Informed by a formative study, the system represents pitch and emphasis together with word and pause durations on a word-aligned canvas. Users can edit these properties through direct controls, natural-language instructions, or vocal demonstrations. Our two-part evaluation examined the methods on assigned tasks and their combined use during open-ended editing. Participants selected methods according to their goals and confidence, then switched to repair results or explore alternatives as new editing needs emerged. Direct manipulation commonly supplied the final adjustment, while earlier instructions and demonstrations contributed to the overall direction. Participants also evaluated the integrated interface positively. Together, the system and findings illustrate how complementary methods can support continued revision of synthesized speech. The implications point out directions for future speech editing tools to consider users' intent expression capabilities, address collateral changes and reuse editing context across operations.

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A Statistics

Table 2. Summary of across-method comparisons in Part 1.

Outcome	DC	VD	NL	Significant pairwise (p_{Holm}, r_{rb})
Perceived control (M)	5.83	4.75	4.25	DC>VD (.029, .80); DC>NL (.023, .88)
Editing time (M, s)	146.46	147.56	248.98	NL>VD (.032, .77); NL>DC (.015, .87)
Latency-adj. time (M, s)	117.65	135.91	198.37	NL>DC (.015, .87)
Iterations (M)	7.96	1.71	4.17	DC>VD (.001, 1.00); DC>NL (.003, .90); NL>VD (.002, 1.00)
NASA-TLX (M)	30.56	35.07	38.82	n.s. ($\chi^2(2)$=1.17, p = .558)
Unique edited words (Mdn)	10.5	34	20.5	VD>DC (.009, .97); NL>DC (.031, .89)

DC = Direct Controls; VD = Vocal Demonstration; NL = Natural-Language Instruction. p_{Holm} from two-sided exact Wilcoxon signed-rank tests; r_{rb} shown as magnitude. n.s. = not significant at α = .05 after Holm correction.

Table 3. Summary of timing-vs-prominence comparisons within DC (families D, O, R).

Measure (Mdn)	Timing	Prom.	Comparison (p_{Holm}, r_{rb})
Word-edit count	10.5	16	Prom.>Timing (.046, .73)
Unique edited words	5.5	5	n.s.
Re-edit intensity	1.87	3.27	Prom.>Timing (.015, .87)
Pitch operations	1.5	8.5	Prom.>Timing (.012, .95)
Emphasis operations	0.5	6.5	Prom.>Timing (.012, .96)
Duration operations	4	2	n.s.
Repeated emphasis	0	3.5	Prom.>Timing (.023, 1.00)
Repeated pitch	0	2.5	Prom.>Timing (.027, .86)
Repeated duration	0.5	1	n.s.

Prom. = prominence tasks. Repeated-parameter values are defined as parameter-specific WEC − UEW. p_{Holm} from two-sided exact Wilcoxon signed-rank tests; r_{rb} shown as magnitude. n.s. = not significant at α = .05 after Holm correction.

B Participant Background

Table 4. Participant demographic background, study mode, spoken English proficiency, audio-editing experience, and prior text-to-speech (TTS) usage frequency (N = 12).

ID	Age	Gender	Setting	Field / Discipline	Spoken English	Audio Exp.	Prior TTS Usage
P1	25–34	M	In-person	Biology	Advanced	None	Occasionally (<1 / mo.)
P2	25–34	W	In-person	Design / UX	Intermediate	None	Tried 1–2 times
P3	25–34	W	In-person	Social Sciences	Highly fluent	Basic	Tried 1–2 times
P4	25–34	W	Online	Design / UX	Intermediate	Basic	Tried 1–2 times
P5	25–34	M	In-person	CS / HCI	Intermediate	None	Tried 1–2 times
P6	25–34	M	Online	Design / UX	Basic	None	Regularly (≥1 / mo.)
P7	18–24	W	In-person	CS / HCI	Intermediate	Intermediate	Occasionally (<1 / mo.)
P8	25–34	M	Online	CS / HCI	Advanced	Basic	Occasionally (<1 / mo.)
P9	25–34	W	In-person	CS / HCI	Intermediate	Basic	Tried 1–2 times
P10	25–34	W	Online	CS / HCI	Intermediate	Basic	Occasionally (<1 / mo.)
P11	25–34	M	In-person	Design / UX	Intermediate	Basic	Occasionally (<1 / mo.)
P12	25–34	M	Online	Social Sciences	Intermediate	None	Occasionally (<1 / mo.)

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